
Fixed Points of AI Revision: An Explicit Review Stage Turns Convergent Self-Refinement into Non-Convergent Expansion

Chenhao Tan and NeuriCo

Abstract

Iterative “ask an AI to review, then revise, then repeat” loops are increasingly used in automated editing and agentic writing pipelines, on the implicit assumption that they eventually *stabilize* at a polished, fixed document. We test this assumption directly. We formalize the loop as a discrete dynamical system $V_{t+1} = \text{REVISE}(V_t, \text{REVIEW}(V_t))$ and ask whether it reaches a fixed point where further iterations produce no changes. Across 12 scientific abstracts, 8 iterations, and 8 experimental conditions, we find that the decoupled two-stage review→revise loop *does not* converge: the per-step normalized edit distance plateaus at a large residual floor (mean final $\Delta \approx 0.38$ versus an approximate-fixed-point threshold of 0.02; one-sample $t = 11.3$, $p = 2 \times 10^{-7}$), and 0% of trajectories reach a fixed point. The cause is the explicit review stage. A single-stage self-refinement baseline on the *same* documents does converge (mean final $\Delta \approx 0.05$, 42% reach an approximate fixed point, exponential-relaxation fit $R^2 = 0.98$), replicating prior work; inserting a freshly generated peer review at every step turns this convergent relaxation into a non-converging expansion process. A never-satisfied reviewer always emits new demands, which the reviser satisfies by adding text: documents inflate $\sim 7\times$, drift semantically from the original (cosine $0.97 \rightarrow 0.77$), and lose factual fidelity (meaning preservation $10 \rightarrow 6.2$ out of 10)—yet isolated quality scores barely move, so the loop *feels* like it is improving while diverging. Running such loops “to stability” is not a reliable stopping strategy.

1 Introduction

Suppose we hand a document to an AI and ask it to *review* the writing, then hand both the document and that review back and ask it to *revise*, and repeat. Does the process ever settle? This is not an idle question: automated editing assistants, agentic writing pipelines, and self-improving document workflows all run some version of this loop, often with an implicit “keep going until it stabilizes” stopping rule. Whether such a loop *terminates*—and whether its limit is a genuinely better document or a degenerate attractor—determines whether it is safe to run to “stability.”

The loop is a dynamical system. Following recent work that views recursive text generation as a map over documents [Wu et al., 2026], we treat a document version V_t as a state and the two-stage operator as $V_{t+1} = F(V_t)$ with $F = \text{REVISE}(\cdot, \text{REVIEW}(\cdot))$. A *fixed point* is a version the loop no longer changes ($V_{t+1} = V_t$), or, more realistically, a soft region where consecutive versions differ by a tiny, stable edit distance. The natural hypothesis—and the folk intuition behind “revise until it stops changing”—is that the loop contracts toward such a fixed point.

The gap. The closest prior work, Wu et al. [2026], studies a *single-stage* self-refine operator (revise directly, with no separate critique) and finds it relaxes cleanly to a soft fixed point. The canonical Self-Refine loop [Madaan et al., 2023] *fuses* critique and revision in one prompt. Neither isolates the effect of a *decoupled, freshly regenerated review at every step*. This matters because a review stage is

not neutral: a reviewer is instructed to *find weaknesses*, and a capable model can always find some. The skeptical literature—self-bias amplification [Xu et al., 2024], feedback friction [Anonymous, 2025], and recursion-induced degradation [Shumailov et al., 2023]—predicts recursion need not be benign, but does not measure the review→revise fixed-point question directly.

Our approach. We run the two-stage loop for $T = 8$ iterations on 12 fresh 2025-era scientific abstracts, logging every version V_t and every review R_t , and measure convergence with the exact metrics of Wu et al. [2026]: normalized Levenshtein edit distance Δ_t , exact and approximate fixed-point rates, exponential-relaxation fits, plus semantic drift and an LLM-as-judge of quality and meaning preservation. Crucially, we run a *single-stage* self-refine baseline on the same documents as a positive control, and ablate seven factors (temperature, cross-model reviewer, review framing, genre, and length guidance) to find what governs convergence.

What we find. The two-stage loop does *not* converge. Per-step change plateaus at a large residual floor (mean final $\Delta \approx 0.38$, an order of magnitude above the 0.02 approximate-fixed-point threshold; $t(11) = 11.3$, $p = 2.2 \times 10^{-7}$), and 0% of trajectories reach a fixed point. The single-stage baseline, on the *same* documents, does converge (final $\Delta \approx 0.05$, 42% approximate fixed points, fit $R^2 = 0.98$)—so the harness detects convergence when it exists, and the explicit review stage is the decisive difference (Cohen’s $d_z \approx -3.3$, the largest effect in the study). The mechanism is runaway growth: a never-satisfied reviewer emits fresh demands every round, the reviser satisfies them by *adding* text, documents inflate $\sim 7\times$ (a 1.5 KB abstract becomes a 10 KB mini-paper), drift from the original (cosine $0.97 \rightarrow 0.77$), and lose fidelity (~ 4 fabricated claims per document)—while isolated-quality judge scores barely move ($7.5 \rightarrow 8.0$), so the loop looks locally like improvement while globally diverging.

Contributions.

- We formalize and directly test the decoupled review→revise loop as a dynamical system, and show it does *not* reach a fixed point: mean final $\Delta \approx 0.38$ and 0% convergence across all two-stage conditions (section 4).
- We identify the explicit review stage as the cause via a same-document single-stage positive control that *does* converge (42% approximate fixed points, $R^2 = 0.98$), the dominant effect in a seven-way ablation ($d_z \approx -3.3$).
- We characterize the failure mechanism—never-satisfied-reviewer-driven length inflation ($\sim 7\times$) that trades meaning preservation ($10 \rightarrow 6.2$) for flat, slightly-rising isolated quality scores—and show it is not a truncation artifact.
- We draw a practical implication: “run until it stabilizes” is unsafe for review→revise loops; a fixed point requires either dropping the review stage or adding an explicit stop criterion.

The rest of the paper reviews related work (section 2), formalizes the operator and setup (section 3), presents results and ablations (section 4), discusses interpretation and limitations (section 5), and concludes (section 6).

2 Related Work

Iterative self-refinement. A model can improve its own output by generating feedback and revising. Self-Refine [Madaan et al., 2023] is the canonical loop, fusing critique and revision in a single prompt; Reflexion [Shinn et al., 2023] accumulates verbal self-feedback across attempts; and CRITIC [Gou et al., 2024] critiques then corrects using *external tools* for the signal. Our operator is a structured variant of this family, but we deliberately *decouple* critique from revision into two separate calls and regenerate the review at every step, so the critique is a first-class object we can log, ablate, and hold responsible for the dynamics.

Recursive generation as a dynamical system. Wu et al. [2026] is the direct precedent: it frames recursive *single-step* self-refinement as “textual relaxation” toward a model-preferred soft fixed point, and defines the metrics we adopt—normalized Levenshtein transition magnitude, exact ($V_{t+1} = V_t$) and approximate ($\Delta_t < 0.02$ and relative word-count change < 0.03 for two consecutive steps) fixed points, and an exponential-relaxation fit $\Delta_t \approx Ae^{-kt} + c$. On ICML abstracts they report 100% approximate convergence and near-perfect fidelity. We replicate this behavior with our single-stage baseline and show that it *breaks* once an explicit review stage is inserted. Related framings cast self-correction as feedback control with stability thresholds [Anonymous, 2026] and report independent

Condition	Mode	Temp	Reviewer	Framing	Genre	Length
D1_main	two-stage	0.0	GPT-4.1 (self)	structured	scientific	preserve
D2_single_stage	single-stage	0.0	—	—	scientific	preserve
D2_temp_default	two-stage	1.0	GPT-4.1	structured	scientific	preserve
D2_cross_model	two-stage	0.0	GPT-4o	structured	scientific	preserve
D2_framing_harsh	two-stage	0.0	GPT-4.1	harsh	scientific	preserve
D2_framing_lenient	two-stage	0.0	GPT-4.1	lenient	scientific	preserve
D2_genre_prose	two-stage	0.0	GPT-4.1	structured	prose	preserve
D2_unconstrained	two-stage	0.0	GPT-4.1	structured	scientific	none

Table 1: The eight experimental conditions: one direct test (D1_main) plus seven ablations. Bold entries mark each ablation’s departure from D1_main. D2_single_stage—the single-stage SELF-REFINE baseline on the same documents—is the positive control for convergence.

evidence of ChatGPT self-convergence [Xylogiannopoulos, 2026], but none isolate a decoupled review loop.

Limits of self-correction. A skeptical camp warns that recursion need not help. Huang et al. [2024] show LLMs often cannot self-correct reasoning without an external signal and can *degrade* results. Xu et al. [2024] show that iterative self-refinement can *amplify* a model’s own bias, inflating self-perceived quality while true quality stalls—which is precisely why we use a judge from a different model generation than the reviser. Anonymous [2025] find that models resist fully incorporating even high-quality feedback, implying residual, non-converging edit pressure. Our results reconcile the two camps: the convergent relaxation of Wu et al. [2026] holds for the single-stage operator, while the skeptics’ failure modes manifest precisely once an explicit critique loop is added.

Model collapse and self-consuming loops. Shumailov et al. [2023] and Alemohammad et al. [2023] study what happens when generative models consume their own outputs, showing loss of diversity and quality (“model collapse,” “model autophagy disorder”). These results are framed for *training* loops; our setting is *inference-time* document revision, but the degenerate-attractor intuition transfers: we observe not variance collapse but its dual, unbounded expansion with semantic drift. Unlike this line of work, we do not fine-tune; we treat recursion as repeated inference and measure its fixed-point behavior directly.

3 Methodology

3.1 The operator and the fixed-point question

We treat a document version V_t as the state of a discrete dynamical system. The two-stage operator applies the same model family in two separate calls:

$$\begin{aligned} R_t &= \text{REVIEW}(V_t) && \text{stage 1: critique the current version,} && (1) \\ V_{t+1} &= \text{REVISE}(V_t, R_t) && \text{stage 2: rewrite given (version + review),} && (2) \end{aligned}$$

so that $V_{t+1} = F(V_t)$ with $F = \text{REVISE}(\cdot, \text{REVIEW}(\cdot))$. We ask whether the trajectory $V_0, V_1, \dots$ reaches a fixed point. Two prompts implement the operator (full text in the released code): a REVIEW prompt (produce a 5–8 bullet critique; do *not* rewrite) and a REVISE prompt (rewrite to address the review, preserve meaning/claims/scope, and “keep roughly the same length”). For each seed document we run $T = 8$ iterations, logging every V_t and R_t .

3.2 Conditions

We test one direct condition (D1_main) and seven ablations (D2_*), summarized in table 1. The pivotal contrast is D2_single_stage, a single-stage SELF-REFINE baseline (revise directly, with no separate review) run on the same documents—our positive control for whether the harness can detect convergence at all. The remaining ablations vary decoding temperature (0 vs. 1.0), the reviewer model (self vs. a cross-model reviewer), review framing (structured, harsh, lenient), genre (scientific abstract vs. general prose), and length guidance (preserve vs. none).

Condition	mean final Δ	clean min $\Delta^\dagger$	% exact FP	% approx FP	fit floor c	fit R^2	mean trunc. versions
D1_main (two-stage)	0.376	0.340	0.00	0.00	0.307	0.61	3.9
D2_single_stage	0.048	0.046	0.08	0.42	0.048	0.98	0.0
D2_temp_default	0.601	0.567	0.00	0.00	0.455	0.27	4.2
D2_cross_model	0.228	0.234	0.00	0.00	0.250	0.79	2.8
D2_framing_harsh	0.451	0.453	0.00	0.00	0.324	0.57	5.2
D2_framing_lenient	0.245	0.218	0.00	0.00	0.244	0.77	2.9
D2_genre_prose	0.275	0.289	0.00	0.00	0.170	0.83	2.4
D2_unconstrained	0.399	0.473	0.00	0.00	0.384	0.52	5.4

Table 2: Convergence outcomes across conditions. The approximate-fixed-point (FP) threshold is 0.02. Every two-stage condition sits an order of magnitude above it and reaches *zero* exact and approximate fixed points; the fit floor c is large and positive (relaxation toward non-zero churn). Only the single-stage baseline converges (bold): it decays to $c \approx 0.05$ with $R^2 = 0.98$ and reaches an approximate fixed point in 42% of documents. † clean min Δ is the smallest per-step Δ over truncation-free steps (robustness check).

3.3 Models, data, and metrics

Models. The reviser/workhorse is GPT-4.1; the independent cross-model reviewer and the LLM-judge are GPT-4O, a different model generation, honoring the “judge $\neq$ reviser” caution against self-bias [Xu et al., 2024]. Embeddings use TEXT-EMBEDDING-3-SMALL. Decoding uses max_tokens= 2048, seed= 42, and temperature as tabled.¹

Data. We use 12 fresh arXiv abstracts spread across 6 categories for the scientific conditions and 10 Wikipedia introductions for the genre ablation. Using fresh 2025-era documents reduces training-data-contamination risk.

Metrics. Mirroring and extending Wu et al. [2026]:

- **Transition magnitude:** per-step normalized Levenshtein distance $\Delta_t = \text{lev}(V_t, V_{t+1}) / \max(|V_t|, |V_{t+1}|)$.
- **Fixed points:** *exact* ($V_{t+1} = V_t$); *approximate* ($\Delta_t < 0.02$ and relative word-count change < 0.03 for two consecutive steps).
- **Relaxation fit:** $\Delta_t \approx Ae^{-kt} + c$; we report the floor c and R^2 .
- **Semantic drift:** $\cos(\text{emb}(V_0), \text{emb}(V_t))$, and the document length trajectory.
- **Quality/fidelity:** an LLM-judge (GPT-4O) scores clarity, conciseness, and technical precision (1–10) at V_0 , V_{mid} , V_{final} , plus meaning preservation and added/dropped-claim counts of V_{final} vs. V_0 .

Truncation control. Runaway length inflation (section 4.3) pushes later versions past the 2048-token cap. We flag a step as “truncated” if either endpoint was cut off (finish_reason==length) and report a robustness metric—the minimum Δ_t over *truncation-free* steps only (“clean min Δ ”)—to show that non-convergence is not a truncation artifact.

Reproducibility. All chat and embedding calls are disk-cached (2076 entries), so the full study re-runs at \$0 and is crash-resumable; a full re-analysis reported 1222 cache hits at an estimated \$0.00. We release raw per-step trajectories, per-trajectory metrics, aggregates, and statistical tests.

4 Results

4.1 The two-stage loop does not converge

table 2 reports convergence outcomes for all eight conditions. Every two-stage condition sits an order of magnitude above the 0.02 approximate-fixed-point threshold and never reaches it: 0% **exact** and 0% **approximate** fixed points across the board. For D1_main, the exponential fit leaves a large positive floor $c \approx 0.31$ —the process relaxes toward a *moving, non-zero* level of churn, not toward zero. The single-stage baseline, by contrast, decays cleanly to $c \approx 0.05$ with $R^2 = 0.98$ and reaches

¹The plan specified a Claude workhorse; only an OPENAI_API_KEY was available in this environment, and a loud-fail policy in the harness prevents silent mocking, so we ran the study on OpenAI models. The mechanism is model-general in principle; cross-provider replication is future work (section 5).

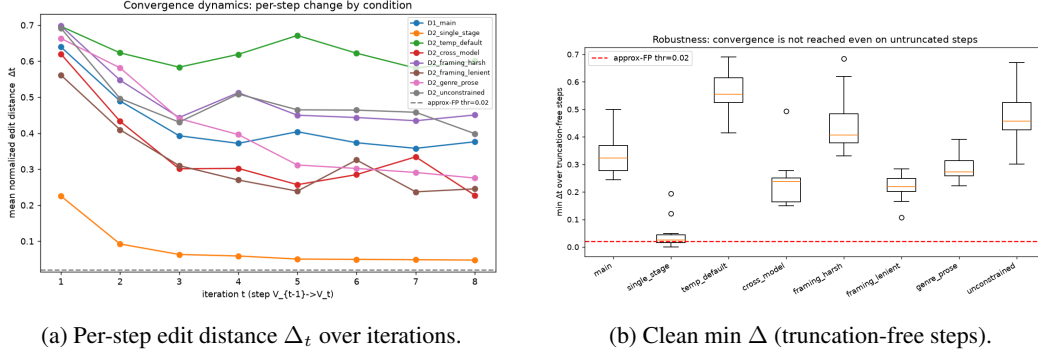


Figure 1: Convergence dynamics. (a) The two-stage REVIEW→REVISE loop (D1_main) plateaus near 0.38; the single-stage SELF-REFINE baseline decays to ≈ 0.05 ; default temperature oscillates without relaxing. The dashed line marks the 0.02 approximate-fixed-point threshold. (b) Even the smallest per-step change over truncation-free steps stays far above the threshold for every two-stage condition, so non-convergence is not a truncation artifact.

Ablation (vs. D1_main)	direction	effect (Cohen’s d_z)	p	reading
single-stage vs. two-stage	↓↓↓ converges	-2.56	5×10^{-4}	the decisive factor
temperature 1.0 vs. 0	↑ more churn / chaotic	+2.09	5×10^{-4}	default temp destroys even relaxation ($R^2=0.27$)
cross-model reviewer vs. self	↓ less churn	-1.17	5×10^{-3}	a different reviewer eases pressure, still no FP
lenient vs. structured framing	↓ less churn	-1.14	7×10^{-3}	less to fix $\Rightarrow$ smaller edits, still no FP
harsh vs. structured framing	↑ (n.s.)	+0.45	0.13	trend toward more churn
unconstrained vs. length-preserving	↑ (n.s.)	+0.19	0.62	length guidance barely helps

Table 3: Determinants of (non-)convergence: paired comparisons against D1_main (Wilcoxon on final Δ). No manipulation produces convergence; the single- vs. two-stage contrast dominates every within-two-stage factor. Negative d_z means less churn than D1_main. n.s. = not significant.

an approximate fixed point in 42% of documents, replicating Wu et al. [2026] and confirming that our harness detects convergence when it exists.

Figure 1a makes the contrast concrete. The two-stage Δ_t curve falls from 0.64 to a plateau near 0.38 by step 3 and stays there; the single-stage curve decays smoothly to ≈ 0.05 ; and at default temperature the curve oscillates in the 0.58–0.70 band with no relaxation at all.

Statistical tests. The result is highly significant. For D1_main, final Δ differs from 0 with $t(11) = 11.30$, $p = 2.2 \times 10^{-7}$, and the truncation-free clean min Δ differs from the 0.02 threshold with $t(11) = 13.01$, $p = 5.1 \times 10^{-8}$ (figure 1b): non-convergence survives the truncation control. The single- vs. two-stage comparison, paired on the same documents, gives Wilcoxon $p = 4.9 \times 10^{-4}$ with **Cohen’s** $d_z = -3.26$ on clean min Δ ($d_z = -2.56$ on final Δ)—the dominant effect in the study.

4.2 Determinants of (non-)convergence

table 3 reports the seven-way ablation. **No manipulation produces convergence.** The dynamics are modulated—a cross-model or lenient reviewer roughly *halves* the residual churn (final Δ from 0.376 to 0.228 and 0.245 respectively), while a higher temperature makes it *worse* and non-monotone (final $\Delta = 0.601$, fit $R^2 = 0.27$)—but the fixed point is never reached. Reviewer identity, framing, and temperature tune the *rate* of churn; only the presence or absence of the review stage changes its *existence*. Length guidance is nearly inert: unconstrained vs. length-preserving is not significant ($p = 0.62$), a point we return to below.

4.3 Mechanism: runaway inflation drives drift, not fidelity

Length explodes. table 4 tracks mean document length from V_0 to V_8 . The single-stage loop holds length ($1.03\times$); every two-stage loop explodes it, with D1_main inflating $6.72\times$ —a 1.5 KB abstract becomes a ~ 10 KB mini-paper (figure 2a). Crucially, the length-preserving instruction (“keep

Condition	V_0	V_8	growth
D2_single_stage	1469	1519	1.03×
D1_main	1469	9869	6.72×
D2_temp_default	1469	9340	6.36×
D2_unconstrained	1469	10478	7.13×
D2_genre_prose	703	10380	14.8×

Table 4: Mean document length (characters), $V_0 \rightarrow V_8$. The single-stage loop holds length; every two-stage loop explodes it. Length-preserving D1_main is indistinguishable from unconstrained D2_unconstrained ($p = 0.62$): review pressure overrides the length instruction.

Condition	$\cos(V_0, V_f)$	mean. /10	+cl.	−cl.
D2_single_stage	0.973	10.0	0.0	0.0
D1_main	0.769	6.2	4.2	2.1
D2_framing_harsh	0.735	4.8	8.4	3.6
D2_genre_prose	0.627	4.8	15.5	1.6

Table 5: Fidelity of V_{final} vs. V_0 : cosine similarity, LLM-judged meaning preservation (/10), and added/dropped claim counts (+cl. / −cl.). Inflation breaks fidelity: two-stage loops drift and fabricate claims, while single-stage preserves meaning perfectly.

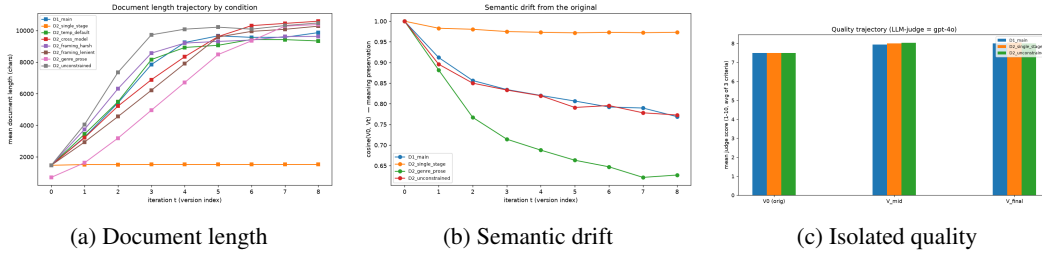


Figure 2: The failure mechanism. (a) Two-stage loops inflate document length $\sim 7\times$ over 8 iterations while the single-stage baseline holds length. (b) Cosine similarity to V_0 falls (from 0.97 single-stage to 0.77 for D1_main): the document drifts from its source. (c) Yet isolated-quality judge scores barely move ($7.5 \rightarrow 8.0$ for D1_main), so the loop looks locally like improvement while globally diverging.

roughly the same length”) is largely *ignored* under review pressure: D1_main (with the instruction) is statistically indistinguishable from D2_unconstrained (without it, $7.13\times$; $p = 0.62$). The review stage’s steady demand for “more precision / examples / discussion” overrides the length constraint.

Inflation breaks fidelity. table 5 shows that inflation, not refinement, dominates. Cosine similarity to V_0 falls from 0.973 (single-stage) to 0.769 (D1_main) and as low as 0.627 (prose; figure 2b), while LLM-judged meaning preservation drops from 10.0 to 6.2 and the loop fabricates ~ 4 new claims per document (up to 15.5 for prose). **Yet isolated-quality scores barely move** (figure 2c): D1_main’s average of clarity, conciseness, and precision goes $7.50 \rightarrow 8.00$ from V_0 to V_{final} —essentially flat and slightly *up*—even as meaning preservation collapses and claims are invented. An LLM-judge scoring documents in isolation is blind to fidelity drift: the loop feels like improvement while diverging from the source, the self-bias / feedback-friction signature.

4.4 Qualitative example

The abstract abs_000 illustrates the mechanism concretely. V_0 (1617 characters) is a dense, single-paragraph arXiv abstract on a “capability-driven data infrastructure” for image generation. V_8 (10479 characters) is a multi-section mini-paper with invented scaffolding—headings such as “Key Concepts and Definitions,” bulleted term glossaries, operational definitions, and examples—none of which were in the original. The review that drove the final step (R_7) reads: “*some definitions . . . could be more precise . . . provide concrete examples for less intuitive terms . . . streamline sections to avoid redundancy.*” Each round the reviewer finds fresh, legitimate-sounding gaps; each round the reviser fills them by *adding* material. There is no state in which the reviewer returns “no changes needed,” so the loop cannot reach a fixed point.

5 Discussion

Answer to the hypothesis. For the two-stage review→revise loop the answer is a clear *no*: it does not converge to a fixed point within a practical horizon; it enters a non-converging *expansion regime*. The hypothesis *is* essentially true for the single-stage variant—self-refine relaxes to a soft fixed point—which is the crucial contrast: the explicit, regenerated review is what prevents convergence, not iteration per se.

Why the review stage matters. A “make this better” editor can declare a document locally optimal and stop editing ($\Delta_t \rightarrow 0$). A “peer reviewer” role is adversarial by construction: it is prompted to *find weaknesses*, and a capable model can always find some. Each critique is fresh edit pressure, and because the cheapest way to “address” most critiques is to add explanatory material, the operator F behaves less like a contraction toward a fixed point and more like a *growth map*. This reconciles the two camps in the literature: the convergent relaxation of Wu et al. [2026] holds for single-stage self-refine, while the skeptics’ warnings—self-bias amplification [Xu et al., 2024], feedback friction [Anonymous, 2025], recursion-induced degradation [Shumailov et al., 2023]—manifest precisely once an explicit critique loop is added. The effect sizes are large and consistent: the single-vs. two-stage contrast ($d_z \approx -2.6$ to -3.3) dwarfs every within-two-stage manipulation.

A caution about isolated LLM-judges. The most striking observation is that absolute quality scores rise slightly while meaning preservation falls sharply. This is a concrete demonstration that an LLM-judge scoring documents *in isolation* is blind to fidelity drift—a caution for anyone using isolated LLM-judge scores as a convergence or quality signal. Fidelity must be measured relative to the source, not judged from the current version alone.

Limitations.

- **Model/provider scope.** We use one reviser family (GPT-4.1) with GPT-4O as cross-model reviewer and judge, due to key availability. The never-satisfied-reviewer mechanism is model-general in principle, but the specific non-convergence should be replicated on Claude, Gemini, and open-weight models before claiming universality.
- **Output-length cap.** The 2048-token cap truncates later two-stage steps (mean ~ 3 –5 of 8 versions in the worst conditions). We mitigate with the truncation-free clean min Δ , which still rejects convergence ($0.34 \gg 0.02$, $p = 5 \times 10^{-8}$); truncation is a *downstream symptom* of inflation, not the cause of non-convergence. A larger cap would give cleaner late-step curves.
- **Horizon.** We cannot rule out convergence at much larger T , but the D1_main Δ_t curve is flat/rising from step ~ 3 and length is still growing at V_8 , so there is no sign of approaching a fixed point.
- **Sample size.** 12 documents per condition (10 for prose) is adequate for the very large primary effects (all $p < 10^{-3}$) but modest for subtle ablation differences (harsh vs. structured is not significant).
- **Judge is an LLM.** Meaning preservation and quality are LLM-judged, not human-annotated; we reduce self-bias by using a different generation than the reviser, but human validation is future work.
- **Prompt dependence.** Results are for one reasonable REVIEW/REVISE prompt pair. A review prompt explicitly permitted to say “no changes needed,” or a reviser with a hard length budget, might converge—a positive-control follow-up.

Broader implications. Iterative review→revise pipelines are deployed under the implicit assumption that they stabilize at a polished document. Our results show the opposite: the loop supplies an endless stream of edit pressure and degrades meaning while its own quality signal reports success. Practitioners should not treat “it stopped changing” as an achievable end state for these loops, and should not trust an isolated quality judge to detect when to stop.

6 Conclusion

Iterating AI *review* → *revise* does *not* reach a fixed point. The regenerated critique injects endless edit pressure; the document inflates $\sim 7\times$, drifts from its original meaning (cosine $0.97 \rightarrow 0.77$), and settles into a high, non-zero churn plateau (mean final $\Delta \approx 0.38$) rather than a stable point, with

0% of trajectories converging. Removing the separate review stage restores convergence to a soft fixed point (42% approximate fixed points, $R^2 = 0.98$), identifying the explicit review loop—not iteration itself—as the cause of non-convergence. Along the way we document a concrete failure of isolated LLM-judges: quality scores rise while meaning preservation collapses.

Takeaway. “Keep asking an AI to review and revise until it stabilizes” is not a reliable stopping strategy; if a fixed point is desired, either drop the separate review stage or add an explicit convergence/stop criterion.

Future work. We highlight five directions: (1) a *positive control for convergence*—give the reviewer an explicit “return NO CHANGES NEEDED” option and/or the reviser a hard length budget, and test whether a fixed point reappears; (2) *cross-provider replication* (Claude, Gemini, open-weight) to establish generality of the never-satisfied-reviewer mechanism; (3) a *length-controlled operator* to disentangle churn from genuine rewriting versus churn from growth; (4) *basins of attraction*—re-seed the same content as paraphrases and test whether trajectories share an attractor; and (5) *human evaluation* of meaning drift to validate the LLM-judge fidelity signal.

References

- Sina Alemohammad, Josue Casco-Rodriguez, Lorenzo Luzi, Ahmed Imtiaz Humayun, Hossein Babaei, Daniel LeJeune, Ali Siahkoobi, and Richard G. Baraniuk. Self-consuming generative models go mad. *arXiv preprint arXiv:2307.01850*, 2023.
- Anonymous. Feedback friction: Llms struggle to fully incorporate external feedback. *arXiv preprint*, 2025.
- Anonymous. Self-correction as feedback control: Error dynamics, stability thresholds, and prompt interventions. *arXiv preprint*, 2026.
- Zhibin Gou, Zhihong Shao, Yeyun Gong, Yelong Shen, Yujia Yang, Nan Duan, and Weizhu Chen. Critic: Large language models can self-correct with tool-interactive critiquing. In *International Conference on Learning Representations (ICLR)*, 2024.
- Jie Huang, Xinyun Chen, Swaroop Mishra, Huaixiu Steven Zheng, Adams Wei Yu, Xinying Song, and Denny Zhou. Large language models cannot self-correct reasoning yet. In *International Conference on Learning Representations (ICLR)*, 2024.
- Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegrefe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, et al. Self-refine: Iterative refinement with self-feedback. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- Ilya Shumailov, Zakhar Shumaylov, Yiren Zhao, Yarin Gal, Nicolas Papernot, and Ross Anderson. The curse of recursion: Training on generated data makes models forget. 2023.
- Wu, Xu, Kang, Chen, Liu, and Yin. Do language models converge to themselves? recursive self-refinement as textual relaxation. *arXiv preprint arXiv:2607.22653*, 2026.
- Wenda Xu, Guanglei Zhu, Xuandong Zhao, Liangming Pan, Lei Li, and William Yang Wang. Pride and prejudice: Llm amplifies self-bias in self-refinement. In *Proceedings of the Association for Computational Linguistics (ACL)*, 2024.
- Konstantinos F. Xylogiannopoulos. Experimental evidence of progressive chatgpt models self-convergence. *arXiv preprint*, 2026.